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Plant Disease Classification

Under the guidance of

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MOTIVATION

India's deepening farm crisis: 76% farmers want to give up farming, shows study

Benefits of government schemes and policies go to big farmers, shows survey undertaken by CSDS on behalf of farmers' association

SOURCE

- Hindustan Times
- Deccan Herald
- Economics Times

Farmers facing repeated crop loss look forward to govt's aid scheme

'Farmers still unaware of schemes'

Hanathe a programme on financial inclusion and literacy comes to a close

Crops worth Rs 50,000 crore are lost a year due to pest, disease: Study

OVERVIEW

- Problem Statement
- Methodology
- Results
- Literature Review
- Implementation
- Future Scope
- Research Gap
- Dataset
- References



PROBLEM STATEMENT

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- Agriculture plays a vital role in feeding the world's population, and plays a major role in a nation's economy.
- But it is constantly challenged by plant diseases that can severely impact crop yield.
- Early detection and identification of plant diseases are critical to ensuring food security and maintaining ecosystem balance.
- Detecting the species accurately can also help us to prevent the disease by following appropriate practises.
- With advancements in deep learning, we can now assist in detecting plant species and their associated diseases. This project aims to use the power of Various Neural Networks to analyze plant images.



LITERARY REVIEW

AUTHOR	METHODOLOGY	LIMITATIONS	ACCURACY
MD. NAHIDUZZAMAN, ABIDA SULTANA, et al..	Lightweight 2D CNN for tomato and cotton leaf disease detection using fewer parameters. Datasets: PlantVillage (tomato)	The model may struggle with new or evolving diseases and relies heavily on high-quality images, limiting real-world scalability.	97.3%
NATASHA NIGAR, MUHAMMAD UMER, et al..	EfficientNetB0, a deep learning model with 237 layers, for classifying 38 plant diseases	Limited dataset diversity and only four models tested; larger datasets and more models could enhance results.	96.89%
R. Singh, A. Jain, et al.	CNN-based cotton leaf disease classification using feature extraction + deep learning	Small dataset size; model may overfit and lacks robustness in field conditions	95.2%
DIANA SUSAN JOSEPH, KAUSTUBH CHAKRADEO, et al..	A CNN model, MRW-CNN is proposed for detecting maize, rice, and wheat leaf diseases.	It does not explore the use of annotated images for advanced object detection models, which could enhance disease detection capabilities.	InceptionV3 on wheat- 94.68%



RESEARCH GAP

● Lack of Diverse Crop Data

Many existing approaches, including lightweight CNN-based models, are designed for specific crops such as tomato and cotton. This restricts their applicability across diverse agricultural scenarios. There is a need for models that can generalize across multiple crop types and disease classes.

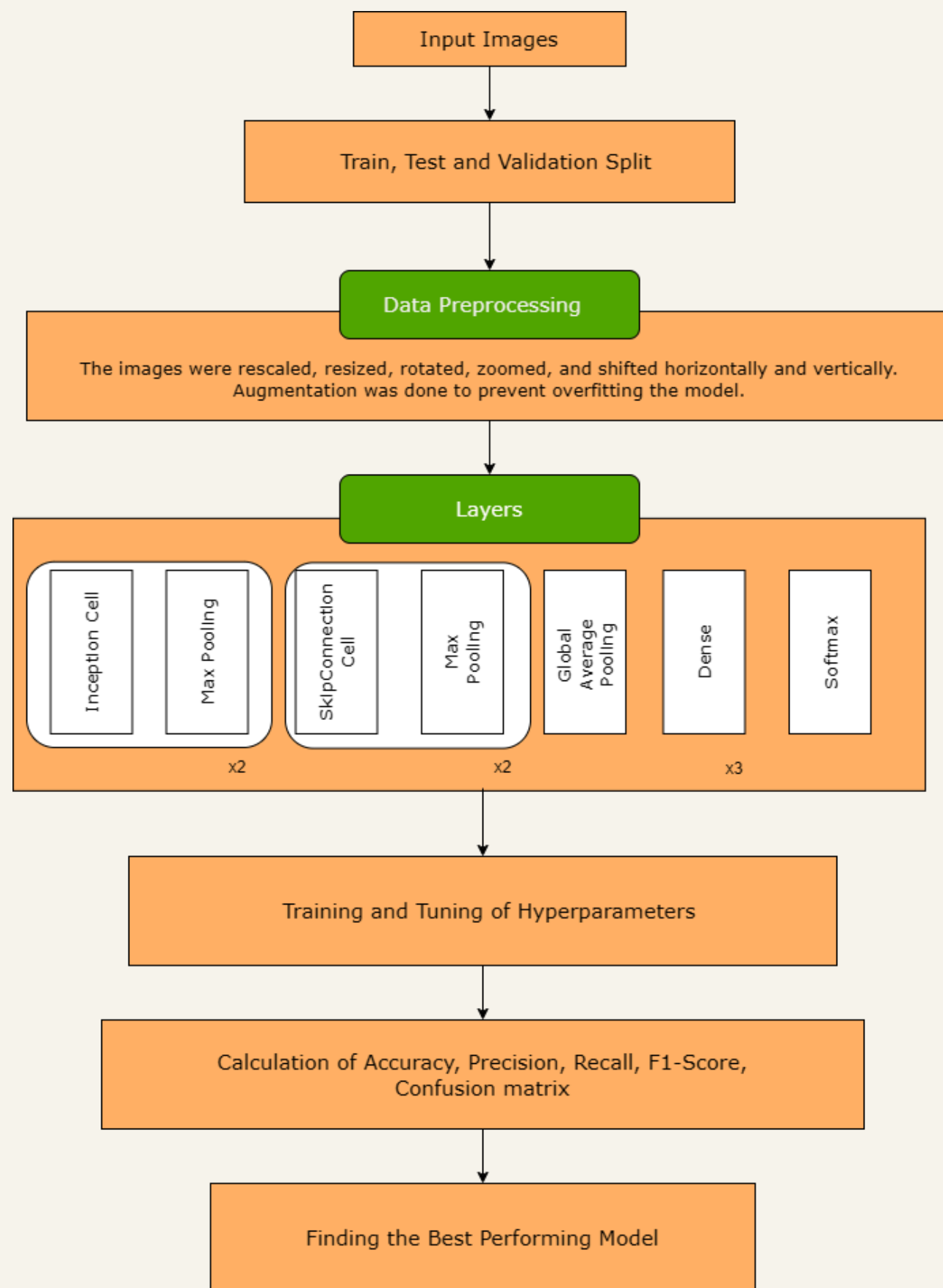
● Existing Architectures

Existing models typically employ either additive (ResNet) or concatenation-based (DenseNet/Inception) feature fusion independently. There is limited exploration of hybrid fusion strategies that can simultaneously preserve information and maintain computational efficiency.



IMPLEMENTATION

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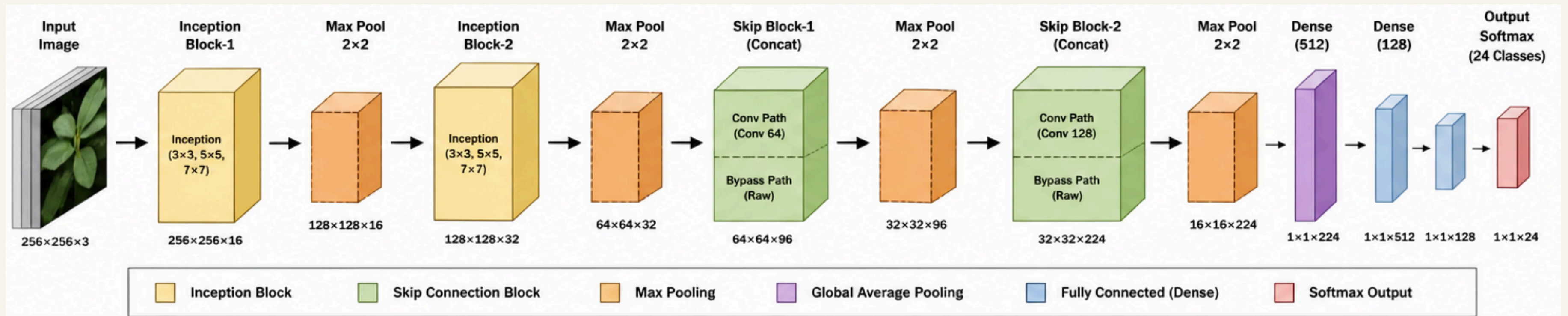
Model Features

Parameters	Values
Input Size	(None, 256, 256, 3)
Batch Size	64
Error Function	Categorical Cross Entropy
Activation	ReLU, Softmax
Optimizer	Adam
Scheduler	Cosine Annealing
Output Size	(None, 24)



IMPLEMENTATION

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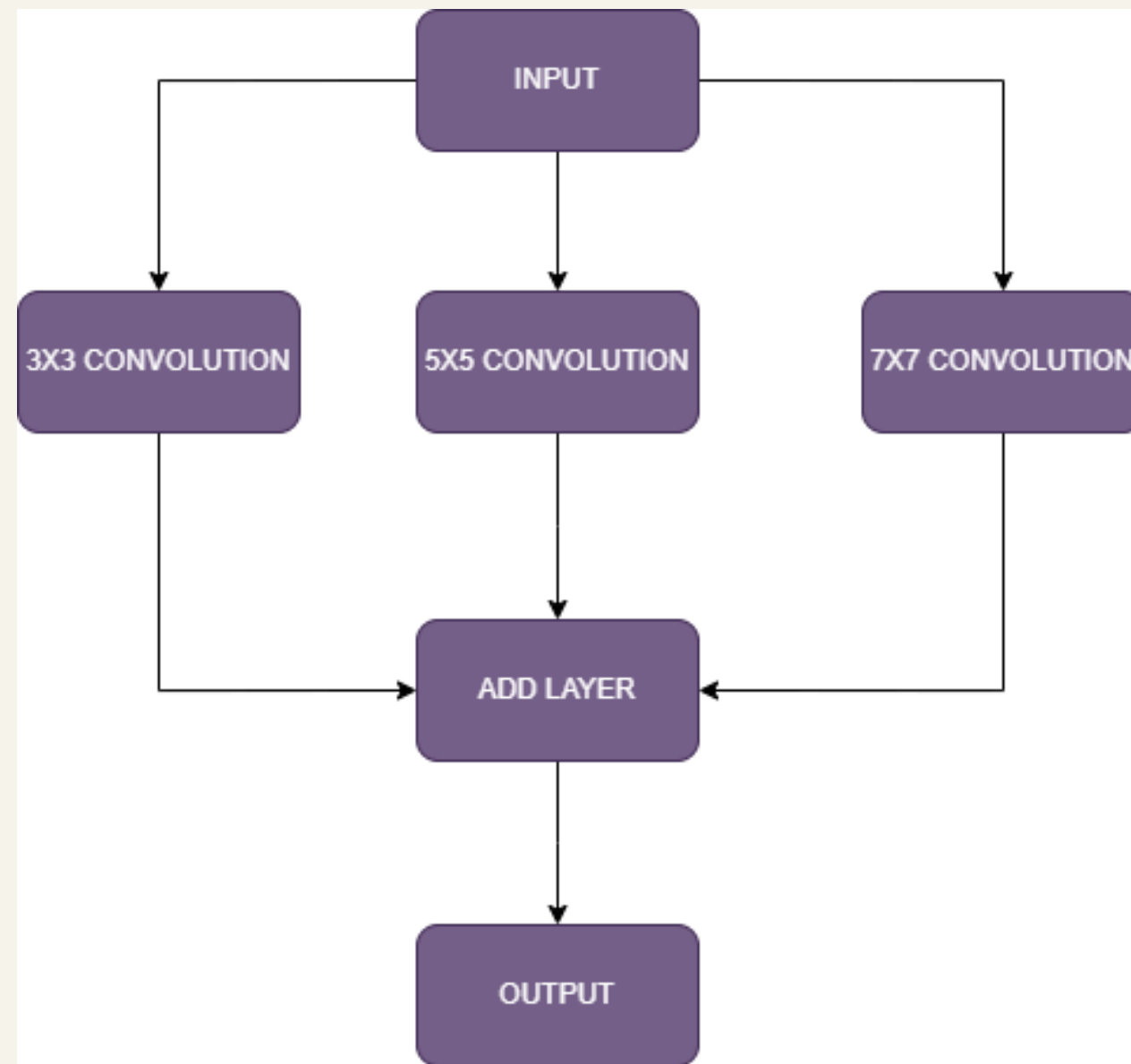




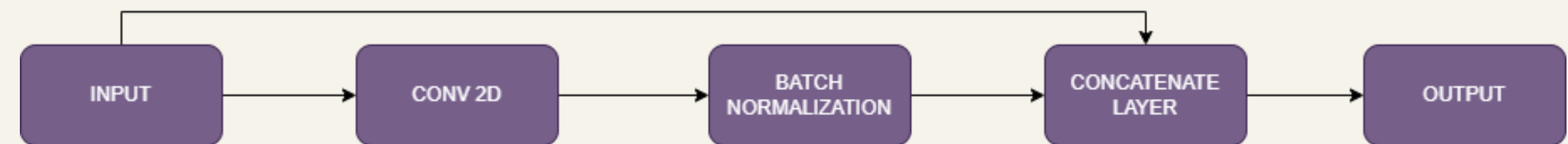
PROPOSED METHODOLOGY

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Inception Cell



Skip Connection Cell





DATASET & TECH STACK

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To develop an effective crop disease detection system, we utilized several key agricultural datasets:

- **PlantVillage Dataset:**

Contains 20,639 images of Bell Pepper, Potato, and Tomato plants, both healthy and diseased. It includes common diseases like late blight and bacterial spot, offering diverse data for accurate disease detection.

- **CottonPlant Disease Dataset:**

Features 4,788 images of Cotton crops, focusing on diseases like bacterial blight and leaf curl. This dataset is essential for cotton-specific disease identification.

- **Wheat Disease Dataset:**

Comprises 2,942 images of Wheat crops affected by diseases such as leaf rust and powdery mildew, supporting precise classification of wheat diseases.

Technologies Used:



Plant	Disease	No of Samples	Plant	Disease	No of Samples
Cotton	Aphids	800	Bell Pepper	Bacterial spot	997
Cotton	Army worm	800	Bell Pepper	Healthy	1478
Cotton	Bacterial Blight	800	Tomato	Bacterial spot	2127
Cotton	Healthy	800	Tomato	Early blight	1000
Cotton	Powdery mildew	800	Tomato	Late blight	1909
Cotton	Target Spot	788	Tomato	Leaf mold	952
Wheat	Brown rust	902	Tomato	Septoria leaf spot	1771
Wheat	Healthy	1116	Tomato	Spider mites	1676
Wheat	Yellow rust	924	Tomato	Target spot	1404
Potato	Early blight	1000	Tomato	Yellowleaf curl virus	3209
Potato	Late blight	1000	Tomato	Mosaic virus	373
Potato	Healthy	152	Tomato	Healthy	1591



RESULTS

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Flip+Brightness Aug
All Datasets

ACCURACY : 0.9706
PRECISION : 0.9707
RECALL : 0.9706
F1-SCORE : 0.9706
SPECIFICITY : 0.9987

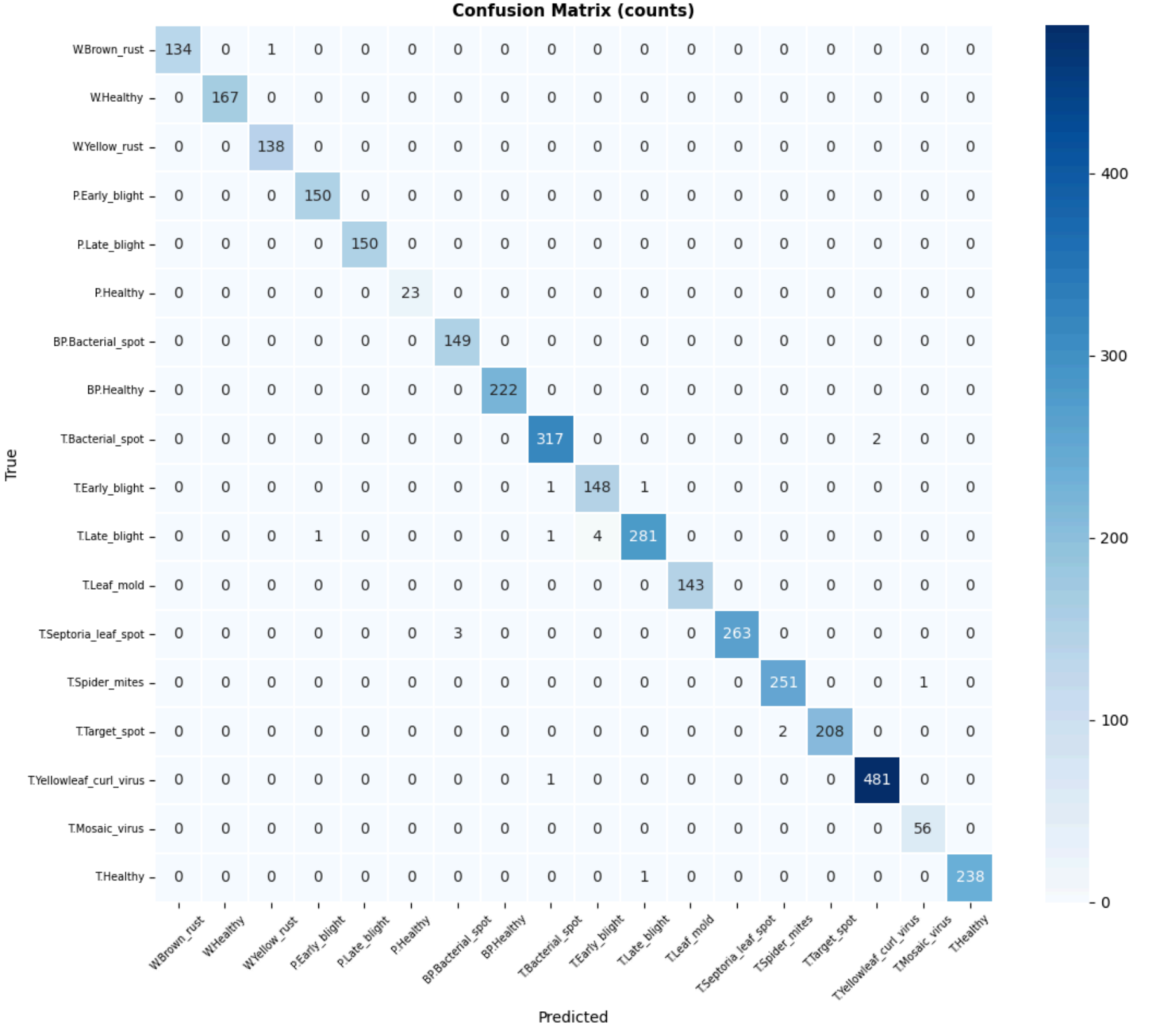
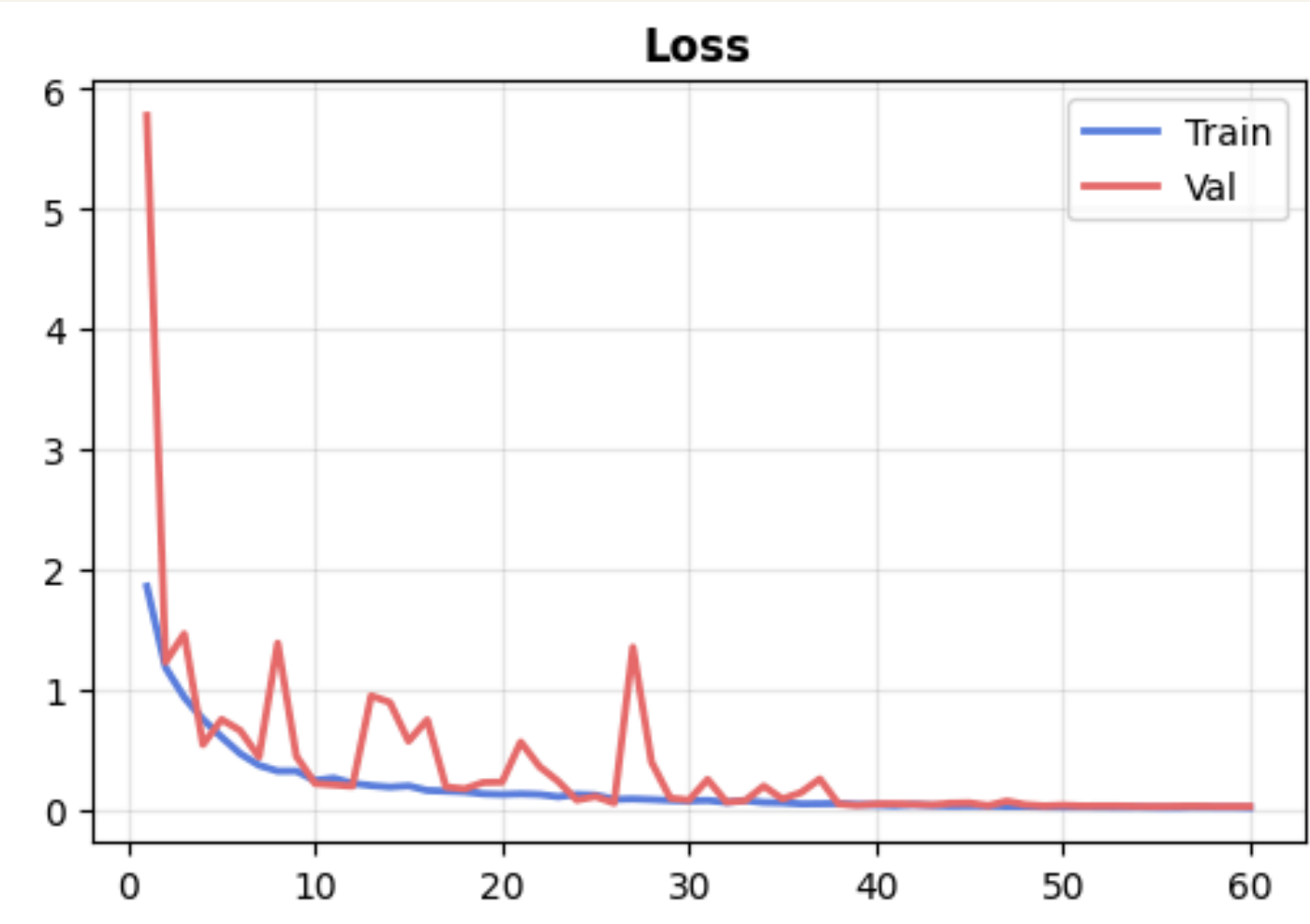
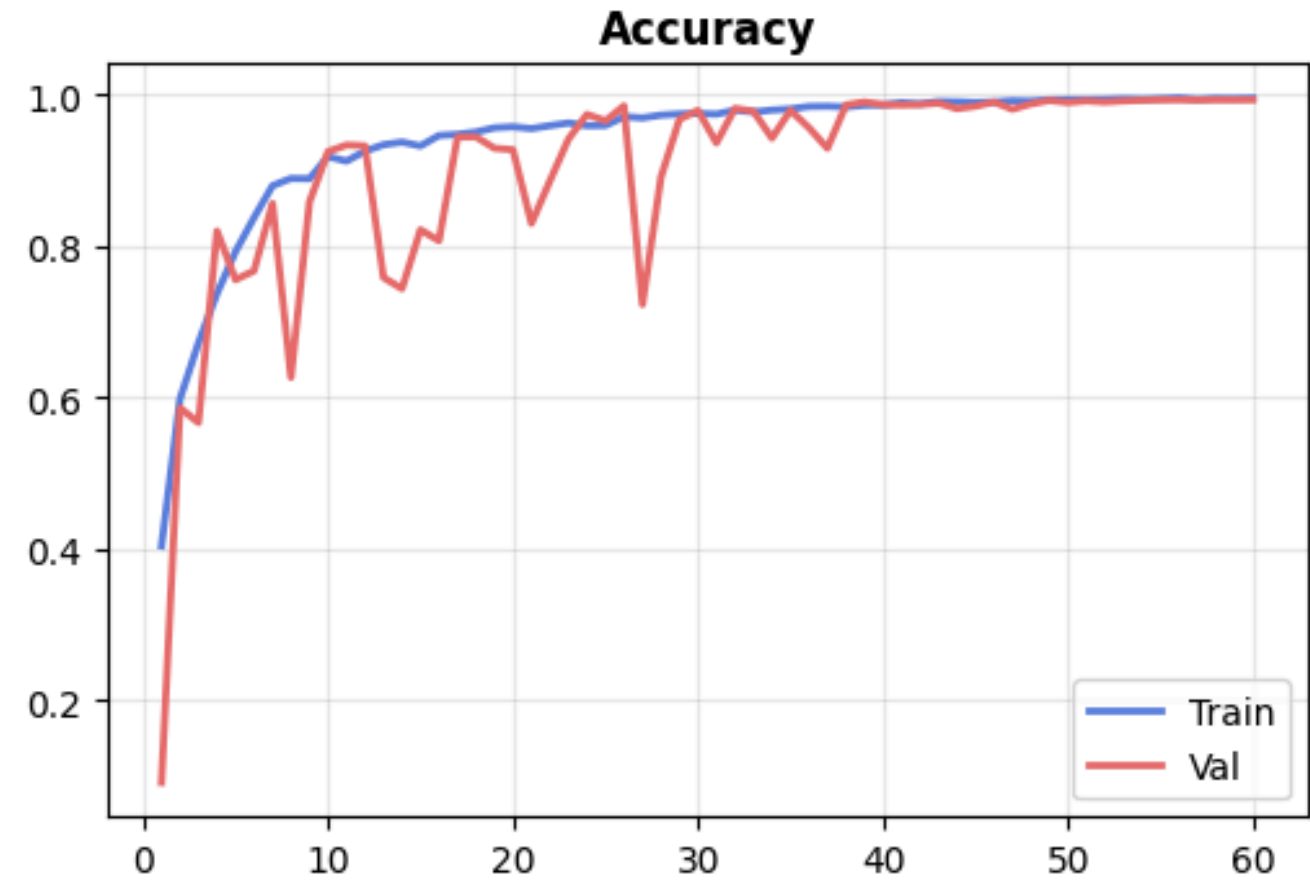
Only PlantVillage,Wheat

ACCURACY : 0.9912
PRECISION : 0.9913
RECALL : 0.9912
F1-SCORE : 0.9912
SPECIFICITY : 0.9995

Metric	Value
Total Parameters	933426
Model File Size (MB)	11.3
FLOPs per image (M)	3984.7 M
Inference single image (ms)	28.06+-4.04
Inference per image in batch (ms)	3.39
Peak memory training (MB)	16.1
Peak memory inference (MB)	0.2
RAM used (MB)	3597.0

		Predicted Class		
		Negative	Positive	
Actual Class	Negative	True Negative (TN)	False Positive (FP) (Type I Error)	Specificity $\frac{TN}{(TN + FP)}$
	Positive	False Negative (FN) (Type II Error)	True Positive (TP)	Recall/Sensitivity $\frac{TP}{(TP + FN)}$
		Negative Predictive Value $\frac{TN}{(TN + FN)}$	Precision $\frac{TP}{(TP + FP)}$	Accuracy $\frac{(TP+TN)}{(TP+FP+TN+FN)}$

RESULTS





FUTURE SCOPE

Future work can focus on advanced regularization and data augmentation techniques to enhance generalization across unseen data while maintaining optimal parameters, fast inference time

The model can be integrated with IoT-based monitoring systems, drones, or automated irrigation systems to enable early disease detection and decision support for precision farming.

Expanding the model to handle multiple crops and real-field conditions will improve robustness and make it more practical for real-world agricultural applications.



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